DEEP VOCAL SOURCE SEPARATION: SPECTROGRAM MASKING, WAVEFORM MODELING, AND HYBRID FUSION ON MUSDB18-HQ

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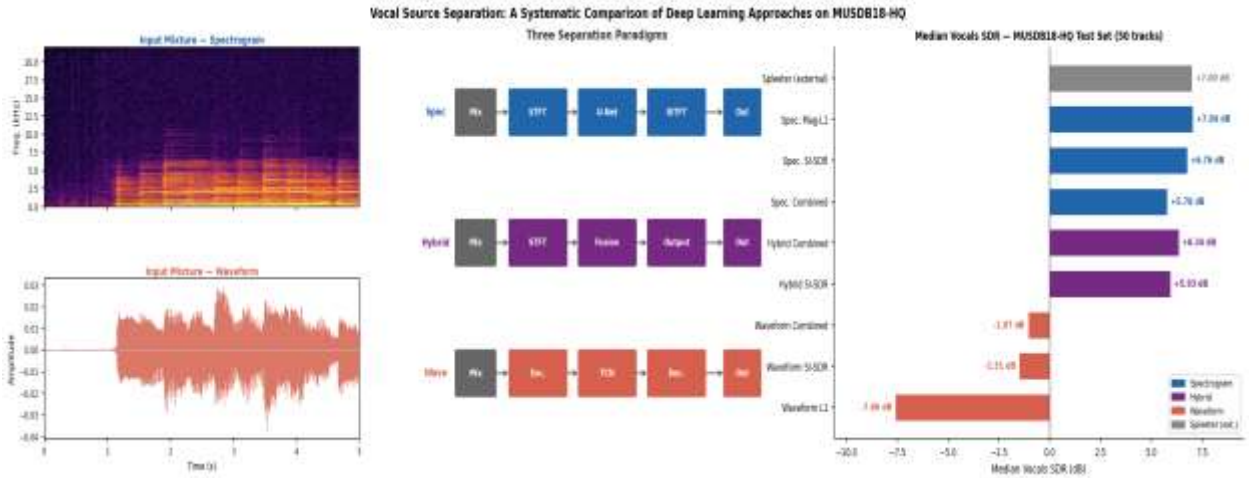
Abstract

Vocal source separation (recovering a clean vocal track from a stereo music mixture) is an ill-posed problem requiring a model to learn what vocals sound like relative to accompaniment from data alone. This work trains and evaluates five deep learning architectures on the MUSDB18-HQ benchmark: a spectrogram-domain U-Net, a waveform-domain Conv-TasNet, a hybrid dual-path model, a Music Demixing Extended network (MDX)-inspired time-frequency variant, and a recurrent spectrogram U-Net. Each architecture is trained under three loss conditions (magnitude L1, scale-invariant signal-to-distortion ratio (SI-SDR), and combined) and evaluated on the official 50-track test set using median vocals signal-to-distortion ratio (SDR), with Spleeter as an external pre-trained reference. The spectrogram U-Net with magnitude L1 loss achieves 7.0 dB median vocals SDR, matching Spleeter at 7.0 dB despite training on only 100 tracks. Waveform models fail to converge competitively under all loss conditions, a gap attributable to memory constraints that forced a reduced sample rate rather than to any inherent limitation of time-domain separation. Results demonstrate that spectrogram masking provides a more data-efficient inductive bias for vocal isolation at this training scale, and that loss function selection is architecture-dependent: objectives suited to one representation degrade severely when applied to the other.



Keywords: *music source separation, vocal isolation, spectrogram masking, U-Net, Conv-TasNet, short-time Fourier transform, permutation-invariant training, MUSDB18-HQ, signal-to-distortion ratio, deep learning*

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Introduction and Problem Statement

Recovering an isolated vocal track from a commercial stereo recording, without access to the original source stems, is a fundamental challenge in audio processing. The task is ill-posed: vocals, drums, bass, and instrumentation are combined into a single stereo signal that can be decomposed in infinitely many ways, requiring a model to learn from data alone what vocals sound like relative to accompaniment. Applications span music production, karaoke generation, lyric alignment, and music information retrieval. This work trains and evaluates deep learning models for two-source vocal separation on MUSDB18-HQ, comparing spectrogram-domain and waveform-domain approaches across multiple architectures and loss functions to understand where the performance gap lies and why. Full four-stem separation and real-time operation are out of scope.

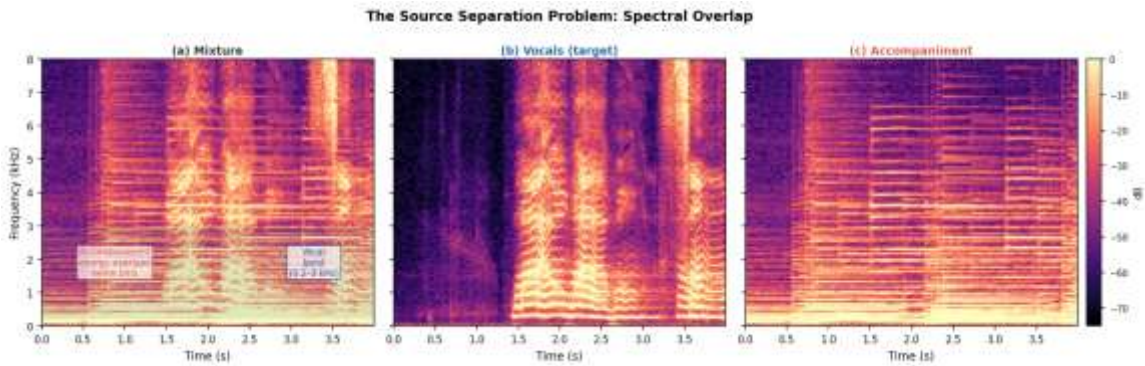

Figure 1. The source separation problem: vocal and accompaniment energy occupy overlapping frequency bins in the mixture (a), making the task ill-posed. The model must recover the vocal stem (b) from a mixture where accompaniment energy (c) saturates the same spectral regions. Excerpt: Lyndsey Ollard: Catching Up, seconds 13–17.

Key Contributions

This work establishes reproducible two-source vocal separation baselines across three model families (spectrogram-domain U-Net, waveform-domain Conv-TasNet, and a hybrid dual-path architecture) under a controlled loss function ablation on MUSDB18-HQ. Spectrogram-domain models achieve up to 7.0 dB median vocal signal-to-distortion ratio (SDR), matching the Spleeter pre-trained baseline at 7.0 dB with no pre-training. Loss function choice proves to be the largest performance differentiator within each architecture family.

Three constraints shaped every design decision. First, 100 training tracks is a small dataset by modern standards; published competitive systems train on thousands of tracks or apply large-

scale data augmentation, neither of which was available here. Second, waveform models operating at 44.1 kHz require substantially more graphics processing unit (GPU) memory than spectrogram models; fitting a competitive waveform model within 24 GB video random access memory (VRAM) required reducing the sample rate to 16 kHz and crop duration to 2 seconds, a known disadvantage relative to published baselines. Third, SDR can penalize a perceptually clean separation when phase reconstruction is imperfect, so headline numbers may understate subjective quality.

The remainder of the report is organized as follows. Section 2 reviews related work on music source separation. Section 3 describes the MUSDB18-HQ dataset. Section 4 details model architectures, loss functions, and training procedures. Section 5 presents evaluation results. Section 6 interprets the findings. Section 7 draws conclusions and outlines directions for future work.

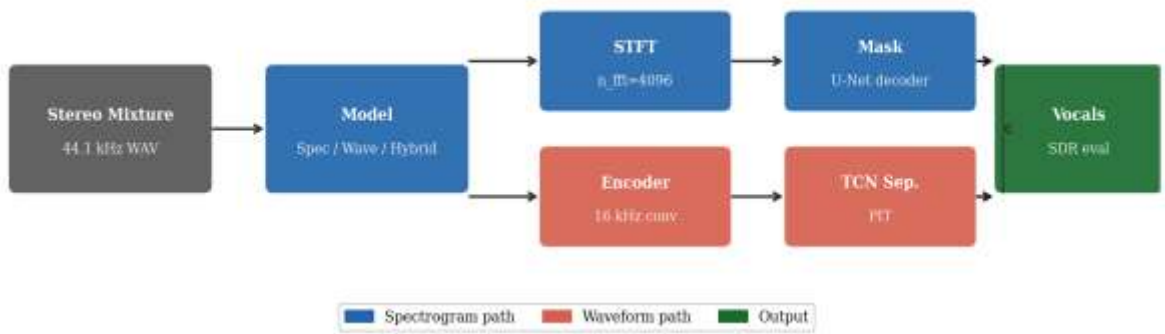

Figure 2. Separation pipeline from stereo mixture input to isolated vocals. The spectrogram path (blue) applies a short-time Fourier transform (STFT) and U-Net mask; the waveform path (red) uses a learned encoder and temporal convolutional network (TCN) separator. Both paths produce a vocals estimate evaluated with mir_eval (a music information retrieval evaluation library) SDR.

Literature Review

Audio Representations for Source Separation

Music source separation presents greater challenges than speech separation: a music mixture contains multiple simultaneous sources with overlapping frequency ranges, diverse timbres, and no reliable formant structure to exploit. Two primary representations of audio are used in

deep learning approaches to the task: the raw waveform and the short-time Fourier transform (STFT) spectrogram.

A waveform is the time-domain representation of a signal, where amplitude is sampled at a fixed rate. Deep learning models that operate directly on waveforms learn their own analysis and synthesis filterbanks without relying on a fixed transform. Luo and Mesgarani (2019) introduced Conv-TasNet, a fully convolutional encoder-decoder network with a temporal convolutional separator, demonstrating that end-to-end waveform models can surpass ideal time-frequency magnitude masks for speech separation. Conv-TasNet has since been widely adopted as a baseline for music source separation tasks.

A spectrogram is produced by applying the STFT to a waveform, yielding a two-dimensional array indexed by time frame and frequency bin. Models typically operate on the magnitude spectrogram, often on a log scale, with phase from the input mixture reused at reconstruction via the inverse STFT. This magnitude-masking approach avoids predicting phase directly, which is difficult to estimate accurately, at the cost of a reconstruction ceiling imposed by phase mismatch. Time-frequency resolution is controlled by two parameters: the window length (n_fft) sets frequency resolution, and the hop length sets temporal resolution. Jansson et al. (2017) applied a U-Net encoder-decoder architecture to log-magnitude spectrograms for singing voice separation, establishing the spectrogram U-Net as a standard baseline for the task. This architecture was later adopted and scaled in Spleeter (Hennequin et al. 2020).

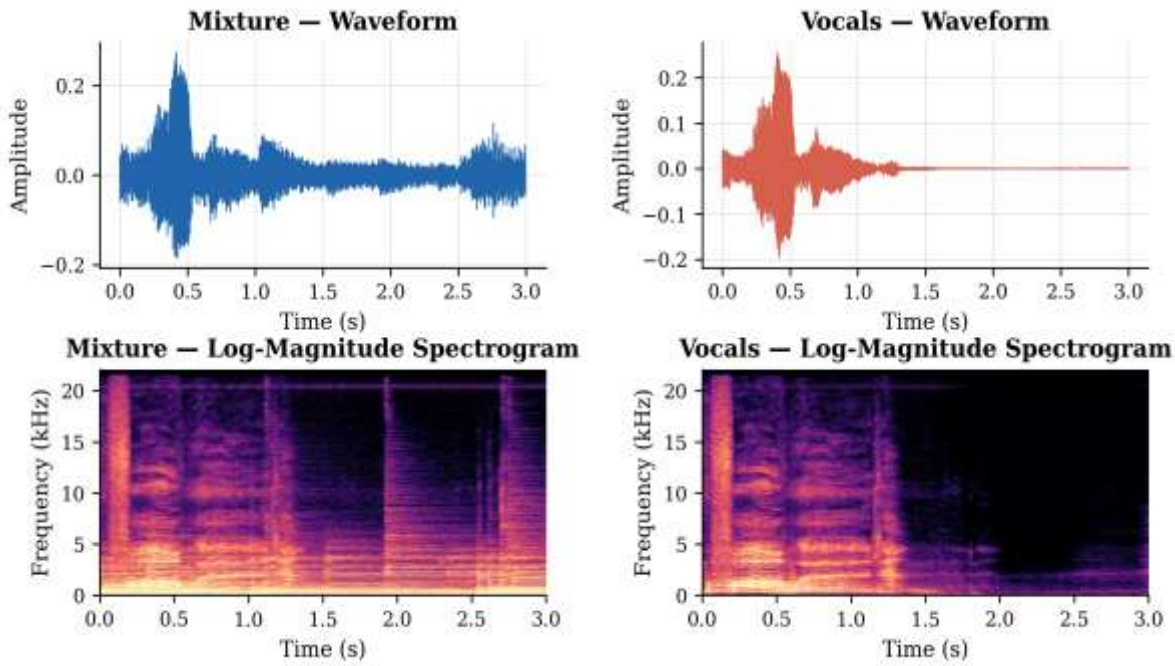

Figure 3. Waveform (top) and log-magnitude spectrogram (bottom) representations of a 3-second excerpt from Lyndsey Ollard: Catching Up. The mixture (left) contains both vocals and accompaniment; the isolated vocals (right) retain only the vocal signal.

Benchmark and Evaluation Standard

MUSDB18-HQ (Rafii et al. 2019) has become the de facto benchmark for supervised music source separation research. It was the official dataset for the SiSEC (Signal Separation Evaluation Campaign) 2018 evaluation campaign (Stöter et al. 2018), and has since been used to evaluate virtually all competitive systems in the field, including Open-Unmix (Stöter et al. 2019), Demucs, and Music Demixing Extended network (MDX)-Net. Its uncompressed, high-quality stems make it particularly suitable for training deep models sensitive to spectral artifacts. The consistent train/test split and standardized evaluation metric (signal-to-distortion ratio (SDR) via mir_eval) allow direct comparison across architectures and loss functions.

Hybrid Architectures

A third line of research combines both representations in a single model, motivated by the complementary strengths and weaknesses of each approach. Spectrogram masking is data-efficient and physically grounded but subject to a reconstruction ceiling: reusing the mixture phase at inference limits separation quality when source and accompaniment phases are

strongly correlated. Waveform models escape this ceiling by operating directly on the signal, but require substantially more training data and compute to learn the structure of audio without any inductive bias from a fixed transform. Hybrid models address this trade-off by running a spectrogram branch and a waveform branch in parallel and combining their outputs through a learned fusion mechanism, allowing the model to draw on the spectrogram prior where it is reliable and the waveform branch where it offers additional flexibility.

Défossez (2021) introduced Hybrid Demucs, a dual-path encoder-decoder that processes the input through both a time-domain waveform branch and a frequency-domain spectrogram branch, with cross-domain feature sharing at multiple encoder depths. Trained on a combination of MUSDB18 and 800 additional proprietary tracks, Hybrid Demucs achieves 7.7 dB median vocals SDR on the MUSDB18 test set, the highest published result on the benchmark at the time and a meaningful improvement over single-representation baselines. The result established the hybrid paradigm as the current state of the art for music source separation and motivated the inclusion of a dual-path hybrid baseline in this work.

Data

MUSDB18-HQ (Rafii et al. 2019) is a curated corpus of 150 full-length stereo music tracks at 44,100 Hz in uncompressed WAV format, partitioned into 100 training tracks and 50 held-out test tracks. Each track supplies four isolated stems (vocals, drums, bass, and other instruments) alongside the mixture, enabling direct supervision of individual sources. The uncompressed stems distinguish it from the original MUSDB18, whose MP3-encoded stems introduce lossy artifacts detrimental to training signal quality. The official train/test split is used throughout without additional data or augmentation. The separation task is defined as two-source: mixture to vocals plus accompaniment, where accompaniment is derived as the mixture minus the vocal stem. Image retrieved from the MUSDB18-HQ dataset (Rafii et al. 2019).

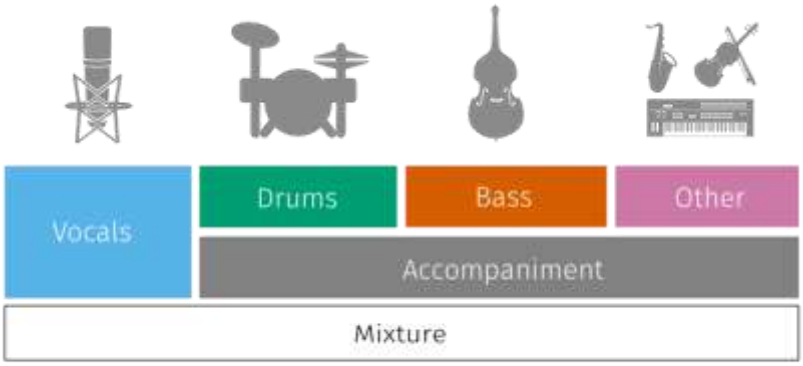

Figure 4. MUSDB18-HQ stem structure. Each track provides four isolated stems (vocals, drums, bass, and other instruments) alongside the stereo mixture. This project uses the two-source formulation: the model separates vocals from accompaniment, where accompaniment is derived as the mixture minus the vocal stem. Image retrieved from the MUSDB18-HQ dataset (Rafii et al. 2019).

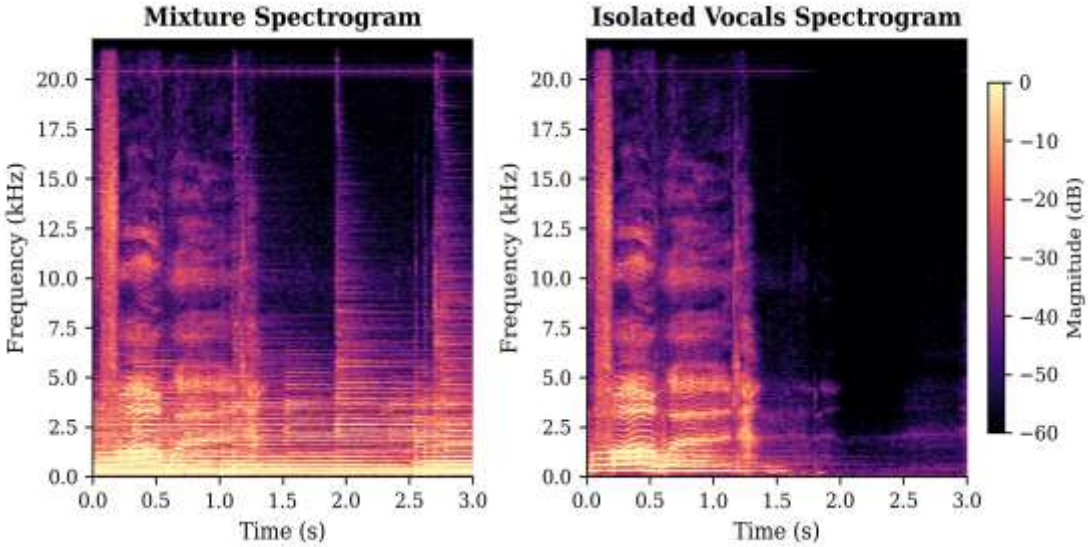

Figure 5. Log-magnitude short-time Fourier transform (STFT) spectrograms of the mixture (left) and isolated vocals (right) for a 3-second excerpt. Vocal energy concentrates in the 0.2–4 kHz band; the mixture spectrogram contains additional energy from percussion and instrumentation across the full frequency range.

Preprocessing

Audio is loaded directly from WAV files without normalization or amplitude scaling. Training examples are random fixed-length crops drawn from each track rather than full tracks, which regularizes training and enables consistent mini-batch construction. Waveform models (Conv-TasNet) use 2-second crops resampled to 16 kHz. Spectrogram and hybrid models use 6-second crops at the native 44,100 Hz. All audio is retained as stereo (2 channels) throughout the pipeline. A validation split of 10% of the training set is held out with a fixed random seed (67), ensuring reproducible splits across all experimental runs. Validation loss drives checkpoint selection and early stopping.

Methods

Evaluation Protocol

Models are evaluated on the official MUSDB18-HQ test set of 50 tracks. Separation quality is measured with Signal-to-Distortion Ratio (SDR), computed per track using mir_eval and aggregated across the test set. Median vocals SDR is the primary metric, with mean included where relevant. SDR captures the ratio of target signal energy to total distortion in the estimate, expressed in decibels. Higher values indicate better separation.

Baseline Methods

The spectrogram baseline applies a five-layer encoder-decoder U-Net to the log-magnitude short-time Fourier transform (STFT) of the stereo mixture. The network predicts a soft magnitude mask, which is multiplied against the mixture magnitude to produce the estimated vocal magnitude. At inference the mixture phase is reused with the estimated magnitude to reconstruct the waveform via the inverse STFT. The STFT uses a fast Fourier transform (FFT) size of 4096, a hop length of 1024, and a Hann window at 44,100 Hz with 6-second training crops. The U-Net has 64 base channels with depth 5.

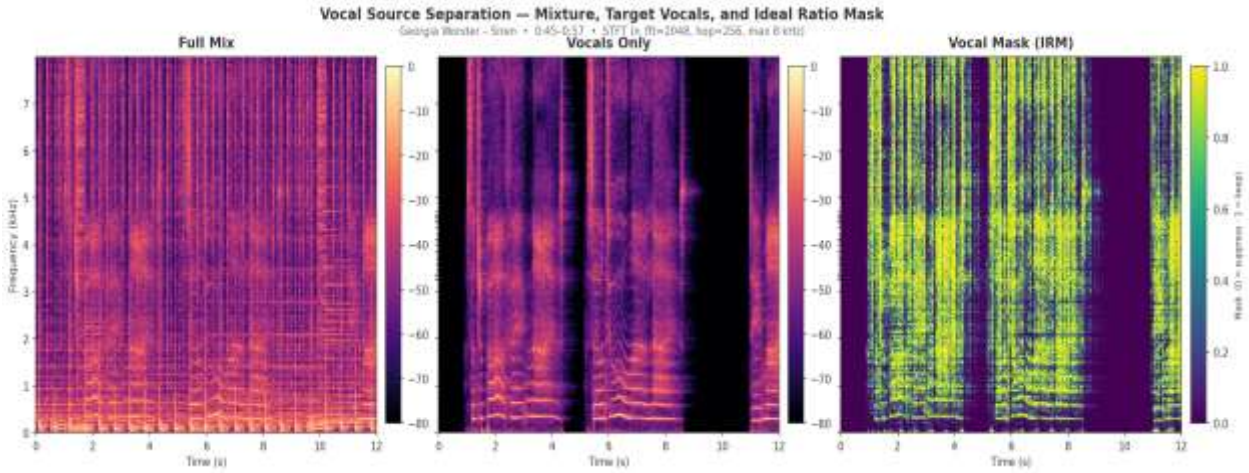

Figure 6. Log-magnitude spectrograms for a 12-second excerpt of Georgia Wonder – Siren (MUSDB18-HQ test set). Left: full stereo mixture. Centre: ground-truth vocal stem. Right: ideal ratio mask (IRM), where values near 1 (yellow) indicate time-frequency bins dominated by vocals. The spectrogram U-Net is trained to approximate this mask.

The waveform baseline uses a Conv-TasNet architecture that operates directly on the time-domain signal. The model consists of a learned 1-D convolutional encoder, a temporal convolutional network (TCN) separator with 8 blocks and 3 repeats per block under global layer normalization, and a learned decoder. The model separates two sources (vocals and accompaniment) simultaneously using a softmax output and is trained with Permutation Invariant Training (permutation-invariant training (PIT)) to resolve source ordering at training time. Audio is resampled to 16 kHz and segmented into 2-second crops.

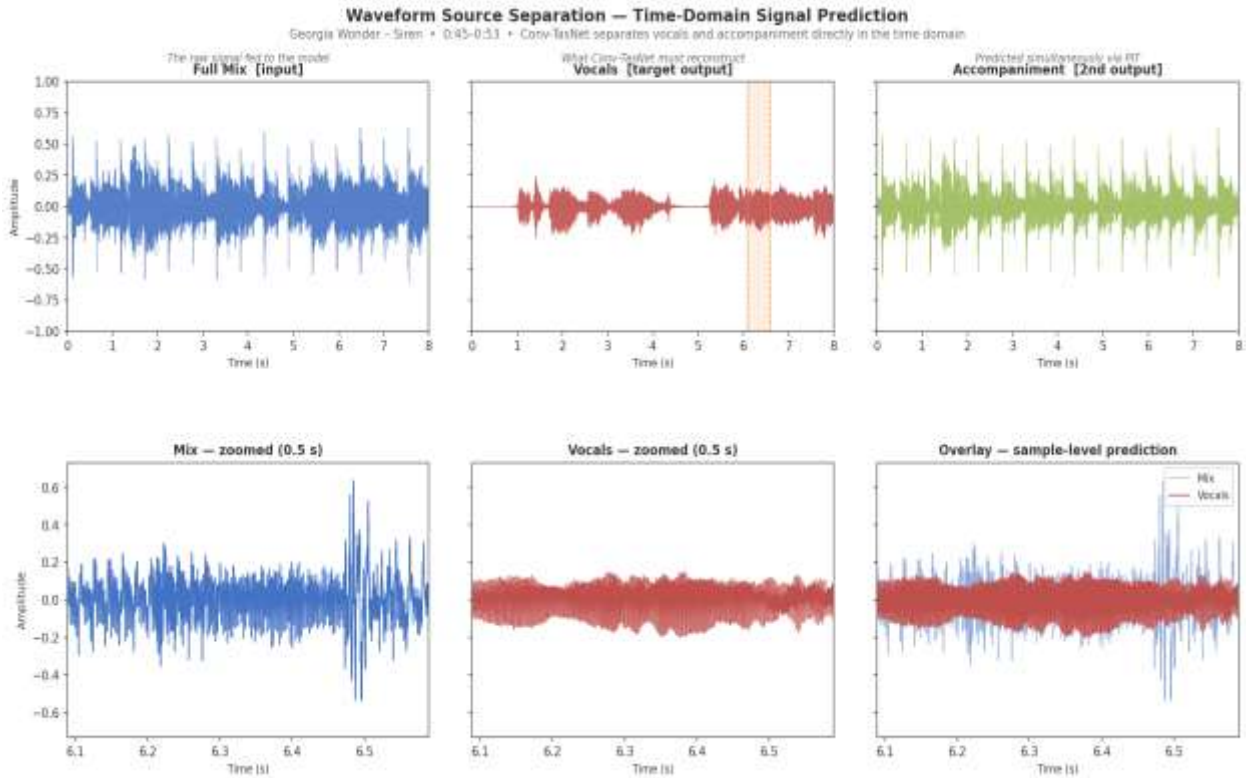

Figure 7. Time-domain waveforms for the same 8-second excerpt (Georgia Wonder – Siren). Top row: full mixture (gray), ground-truth vocal stem (orange), and accompaniment (blue) as seen by Conv-TasNet. Bottom row: a 0.5-second zoom at the loudest vocal moment. The overlay shows that the vocal signal is embedded within the mixture at the sample level; the model must reconstruct the exact orange waveform with no frequency-domain mask as an intermediate step.

The hybrid baseline combines both representations in a dual-path architecture. A spectrogram branch applies a six-layer U-Net to the log-magnitude STFT (n_fft 4096, hop 1024) to produce a magnitude mask. A waveform branch processes the time-domain signal through a Conv-TasNet-style encoder and separator (8 blocks, 2 repeats, 256 channels). The outputs of both branches are fused with a learned gated fusion layer. The model runs at 44,100 Hz with 4-second training crops. Under the combined loss condition, a weighted sum of scale-invariant signal-to-distortion ratio (SI-SDR) (weight 1.0) and multi-resolution STFT loss (weight 0.1) is applied to the fused waveform output; no branch-specific loss terms are used.

Figure 8. Architecture overview. Top: Spectrogram U-Net applies an STFT, predicts a soft magnitude mask through a 5-layer encoder-decoder, and reconstructs via inverse short-time Fourier transform (ISTFT) with mixture phase reuse. Middle: Hybrid dual-path model runs a spectrogram branch and a waveform branch in parallel, combines their features via gated fusion, and decodes with a combined loss. Bottom: Waveform Conv-TasNet encodes the raw signal with learned basis functions, separates two sources with a TCN using PIT, and decodes with a transposed convolution.

Loss Function Ablation

To examine the effect of the training objective, each baseline is trained under three loss configurations while holding architecture and data fixed. The three conditions are L1 magnitude loss (mag L1), scale-invariant SDR (SI-SDR), and a combined loss. For the spectrogram model, the combined loss adds SI-SDR (applied after inverse STFT reconstruction) and a multi-resolution STFT loss evaluated at three scales (FFT sizes 2048, 1024, and 512). For the waveform model, the combined loss adds SI-SDR and the same multi-resolution STFT loss. Ablation results are reported in the Results section.

Additional Models

Two further model variants are included beyond the baselines. The first is a Music Demixing Extended network (MDX)-inspired spectrogram model that replaces the standard U-Net convolutional blocks with time-frequency dense (TDF) blocks, each consisting of a 3x3 convolution followed by 1x1 channel mixing. The MDX model uses the same STFT parameters, sample rate, and training configuration as the spectrogram baseline. The second is a recurrent spectrogram model that augments the U-Net baseline with a bidirectional GRU layer (hidden size 128) at the network bottleneck to capture longer-range temporal context, with all other architecture and training settings unchanged.

Training

All models are trained with the Adam optimizer. The checkpoint with the lowest validation loss is retained, and training terminates after 20 consecutive epochs without improvement. Waveform models use a lower learning rate and gradient clipping to stabilize training under the permutation-invariant objective. All training is conducted on a single NVIDIA RTX 4090. Full hyperparameter details for each model family are provided in Appendix A.

Results

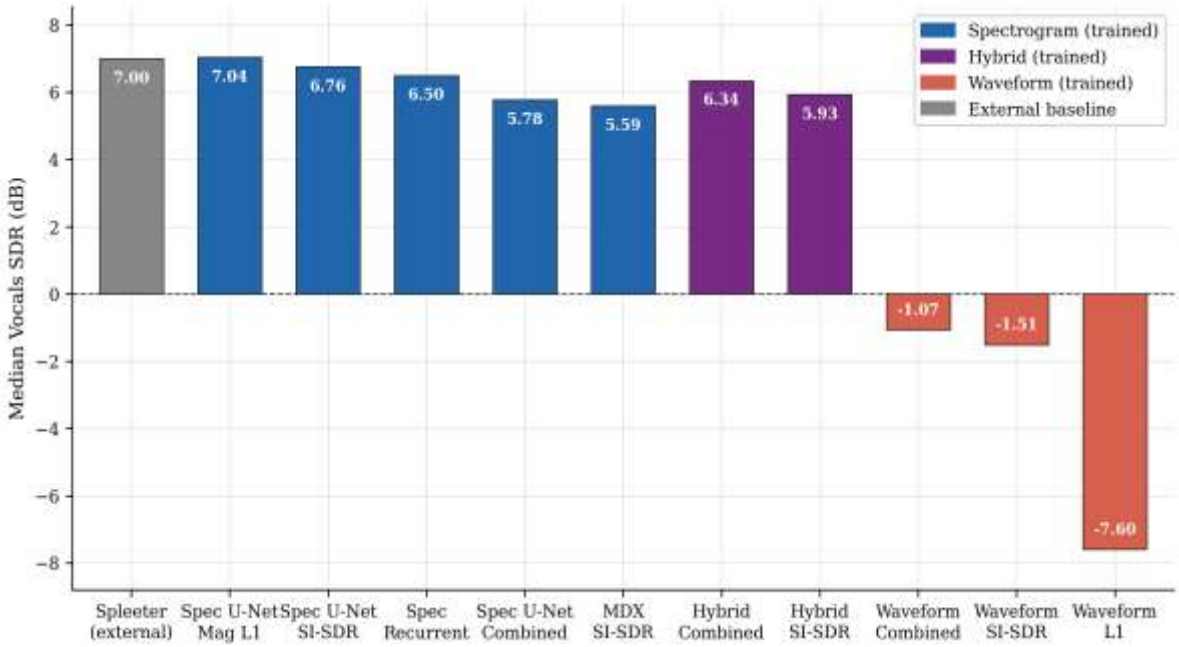

Figure 9. Median Vocals signal-to-distortion ratio (SDR) (dB) for all evaluated models on the MUSDB18-HQ test set (50 tracks). Colors denote model family: blue = spectrogram, purple = hybrid, red = waveform, grey = external baseline.

Table 1 and Figure 9 summarize median vocals SDR across all evaluated models on the 50-track MUSDB18-HQ test set. Spectrogram models cluster between 5.59 and 7.0 dB, with the U-Net magnitude L1 configuration at the top and the external Spleeter baseline at 7.0 dB sitting within 0.0 dB of it. This narrow gap is notable: Spleeter was trained on a large proprietary dataset, while the spectrogram U-Net was trained only on the 100 MUSDB18-HQ training tracks. Waveform models occupy a separate performance tier entirely, reaching a best of -1.1 dB under combined loss, more than 8 dB below the weakest spectrogram configuration. The 1.3 dB spread across all spectrogram conditions is smaller than the gap between any spectrogram model and any waveform model, indicating that representation choice is the dominant factor at this training scale.

Median Vocals SDR — All Models, MUSDB18-HQ Test Set (50 tracks)

Model	Loss	Median Vocals SDR (dB)
Spectrogram U-Net	Mag L1	+7.04
Spleeter (external)	—	+7.00
Spectrogram U-Net	SI-SDR	+6.76
Spec Recurrent	SI-SDR	+6.50
Hybrid	Combined	+6.34
Hybrid	SI-SDR	+5.93
Spectrogram U-Net	Combined	+5.78
MDX	SI-SDR	+5.59
Waveform Conv-TasNet	Combined	-1.07
Waveform Conv-TasNet	SI-SDR	-1.51
Waveform Conv-TasNet	L1	-7.60

Table 1. Median vocals SDR for all evaluated models. Color bars on the left indicate model family. SDR values shown in green (positive) and red (negative).

Loss Function Ablation

Table 2 isolates the effect of training objective, holding architecture and data constant. For the spectrogram U-Net, all three loss conditions produce positive, competitive results, spanning a 1.3 dB range from 7.0 dB (magnitude L1) to 5.8 dB (combined). For the waveform Conv-TasNet, all three conditions produce negative median SDR across the test set, with a 6.5 dB spread from -1.1 dB (combined) to -7.6 dB (L1). Two patterns stand out. First, the loss that performs best for the spectrogram model (magnitude L1) produces catastrophic failure for the waveform model, confirming that loss function selection is architecture-specific. Second, the combined loss improves waveform performance by 6.5 dB relative to L1 while slightly degrading spectrogram performance by 1.3 dB, a divergence that reflects the fundamental difference in what each architecture predicts.

Loss Function Ablation — Same Architecture, Different Loss

Architecture	Loss	Median Vocals SDR (dB)
Spectrogram U-Net		
Mag L1		+7.04
SI-SDR		+6.76
Combined		+5.78
Waveform Conv-TasNet		
Combined		-1.07
SI-SDR		-1.51
L1		-7.60

Table 2. Loss function ablation results. Same architecture trained under three loss conditions for spectrogram U-Net and waveform Conv-TasNet.

Per-Track Variance

Per-track vocals SDR varies substantially across the 50 test tracks (Figure 10). Under the best spectrogram model, the interquartile range spans 4.26 to 8.6 dB (IQR 4.4 dB), with 6% of tracks yielding negative SDR. The best waveform model shows a compressed distribution centered well below zero (Q1 -2.7 dB, Q3 1.0 dB), with 58% of tracks yielding negative SDR. The highest-scoring tracks under the spectrogram model are Lyndsey Ollard - Catching Up (13.7 dB) and Sambasevam Shanmugam - Kaathaadi (12.1 dB). The lowest are PR - Happy Daze (-23.9 dB) and Skelpolu - Resurrection (-10.7 dB). Notably, these failure tracks rank last across every model family tested, suggesting that failure is driven by acoustic properties of the source material rather than by any particular architecture. Full per-track scores for all models are reported in Appendix B.

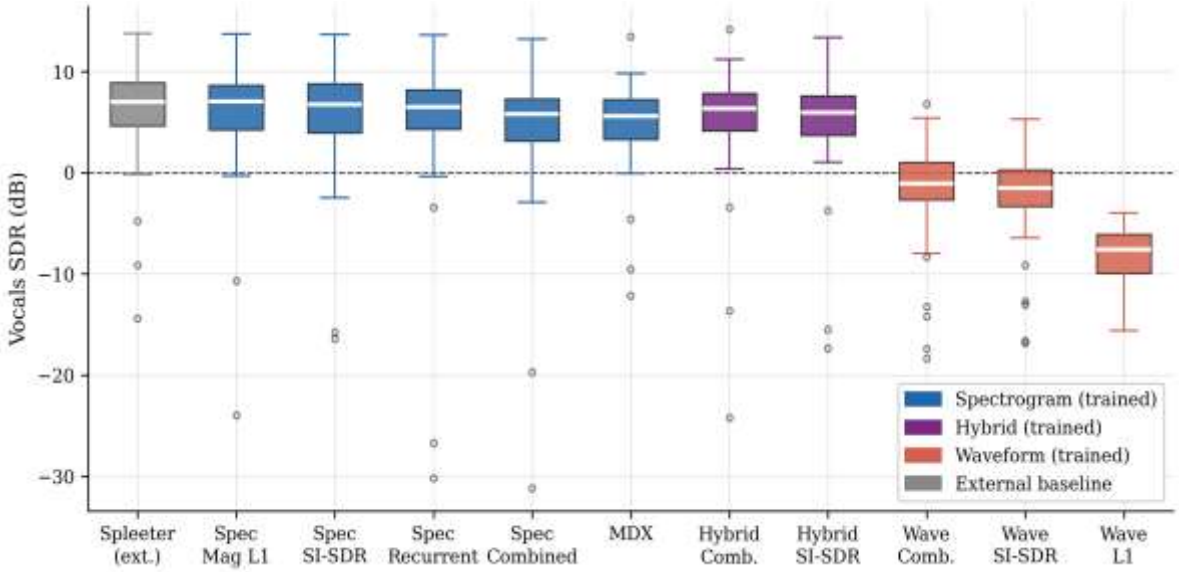

Figure 10. Per-track vocals SDR distribution for all models on the MUSDB18-HQ test set (50 tracks). Boxes show the interquartile range, the white line marks the median, and whiskers extend to 1.5x IQR.

Discussion

Spectrogram Models Outperform Waveform Models by a Wide Margin

Spectrogram and waveform models differ by more than 8 dB in median vocals SDR. The best spectrogram model achieves 7.0 dB median vocals SDR, while the best waveform model reaches only -1.1 dB under the combined loss, a difference of more than 8 dB. Only 6% of test tracks yield a negative SDR under the best spectrogram model, compared to 58% under the best waveform model. This gap is not a consequence of architecture capacity alone. The performance gap is primarily driven by the inductive bias of the spectrogram representation. The spectrogram U-Net applies a soft magnitude mask to the mixture spectrogram and reuses the mixture phase at inference. This constrains the output to a filtered version of the input, which is physically reasonable for source separation and prevents the model from generating energy outside the mixture. End-to-end waveform models carry no such constraint. Conv-TasNet must learn to synthesize plausible source waveforms from scratch, a harder task that requires substantially more training data and capacity than the 100-track MUSDB18-HQ training set provides. The scale of this gap is implementation-specific, driven by the memory constraints detailed in the following subsection, not by any inherent ceiling of time-domain separation.

Waveform Performance Gap Relative to Published Models

The best waveform result achieved here (-1.1 dB) falls well short of published Conv-TasNet variants trained on MUSDB18, but this reflects implementation constraints rather than a ceiling on waveform separation as an approach. Défossez et al. (2019) report a vocals SDR of approximately 6.3–6.8 dB using a comparable waveform architecture, attributing performance to three conditions: native 44.1 kHz sample rate, data augmentation (pitch and tempo shifting, contributing approximately 0.5 dB alone), and extended context windows of 10 or more seconds per training example. Training at 16 kHz with 2-second crops was a deliberate practical choice rather than an oversight. An initial waveform configuration at 44.1 kHz and 6-second crops was attempted on the RTX 4090 (24 GB video random access memory (VRAM)), but the sequence length at native rate exhausted available graphics processing unit (GPU) memory, making stable training infeasible. The down sampled configuration not only fit in memory but also produced a higher validation SDR than the 44.1 kHz run reached before it was halted, suggesting that stable convergence at reduced resolution outperformed unstable training at full resolution given the same hardware constraints. This trade-off illustrates a practical tension in time-domain separation: waveform models benefit from high sample rates but impose memory costs that scale quadratically with sequence length, whereas spectrogram models remain tractable at native rate through short-time Fourier transform (STFT) compression. Closing the waveform gap on consumer hardware would require gradient checkpointing, mixed-precision training, or smaller batch sizes at full resolution, directions identified for future work.

Waveform L1 Loss Is Ill-Suited for Time-Domain Separation

The waveform L1 condition produces a median vocals SDR of −7.6 dB, with 100% of test tracks yielding negative SDR. SDR is expressed in decibels: positive values indicate that the target signal dominates the estimate, negative values indicate that distortion dominates, and 0 dB represents equal energy between target signal and artifact. A negative SDR is therefore worse than outputting silence.

This result is not a poorly tuned model but a fundamental mismatch between loss function and architecture. L1 loss applied directly to raw waveforms penalizes sample-level amplitude differences without any notion of perceptual or spectral structure. A predicted waveform that is slightly time-shifted from the target, even by a fraction of a millisecond, incurs high L1 loss even if it sounds perceptually correct (Défossez et al. 2019). This creates a degenerate training signal: the model has no path to a low-loss solution without perfectly aligning every sample,

which is unachievable from a mixture. The model instead learns to minimize loss by collapsing toward low-energy outputs rather than learning meaningful separation. The finding confirms that waveform models require loss functions invariant to small temporal misalignments, such as SI-SDR or multi-resolution STFT loss (Luo and Mesgarani 2019).

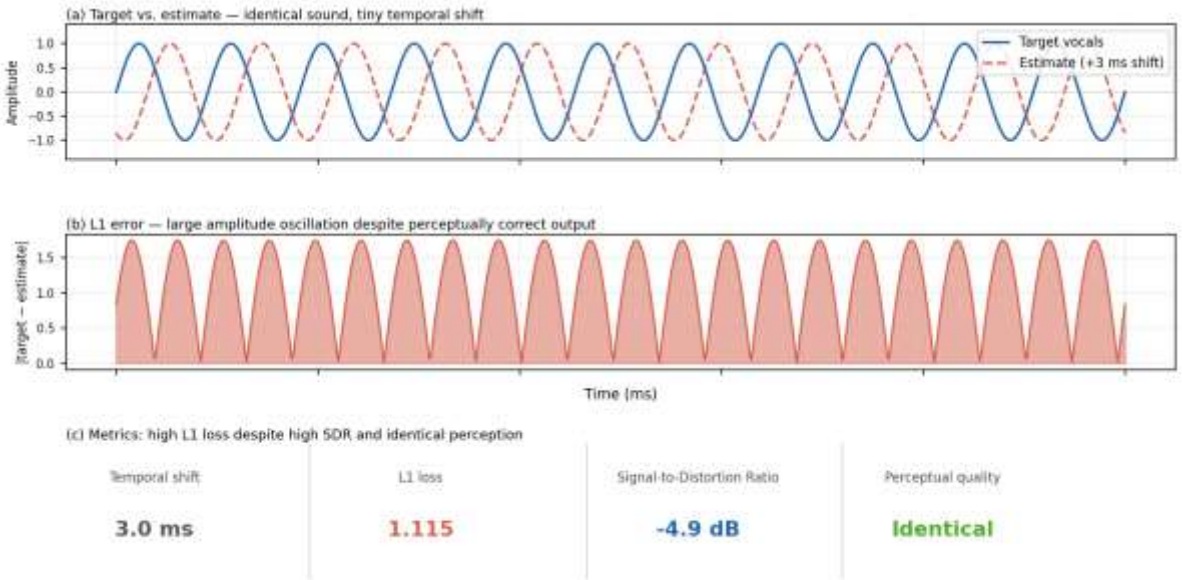

Figure 11. Waveform L1 loss failure mode illustrated with a synthetic 220 Hz sinusoid. A 3 ms temporal shift (panel a) produces an L1 loss of 1.115 and a Signal-to-Distortion Ratio of −4.9 dB (panel c), yet the estimate is perceptually indistinguishable from the target. The L1 error (panel b) oscillates at full amplitude because the sample-level difference between two time-shifted sinusoids is itself a sinusoid. Any realistic waveform model output will be misaligned at the sample level; L1 loss provides no useful gradient signal under these conditions.

Effect of Loss Function on Spectrogram Models

Among the three loss conditions tested on the spectrogram U-Net, magnitude L1 loss produces the best result (7.0 dB), with scale-invariant signal-to-distortion ratio (SI-SDR) close behind (6.8 dB) and combined loss third (5.8 dB). The advantage of magnitude L1 is architectural: the model predicts a magnitude mask, so optimizing the L1 distance between predicted and target magnitude directly aligns the loss with what the model controls. SI-SDR requires reconstructing a waveform via inverse STFT before computing the loss, introducing the mixture phase into the gradient signal and creating a mismatch between what the model can optimize (the mask) and what the loss penalizes (the full waveform). The combined loss compounds this by adding a multi-resolution STFT term, producing conflicting gradients that degrade performance further.

The takeaway is that loss function design must match the model's output domain: a mask-based model benefits from a mask-domain loss.

Hybrid and Additional Models

The hybrid model achieves 6.3 dB under combined loss and 5.9 dB under SI-SDR, placing it between the spectrogram and waveform baselines. The combined loss helping the hybrid while hurting the spectrogram-only baseline suggests that the waveform branch benefits from a time-domain loss signal. The gap between the hybrid (6.3 dB) and the best spectrogram model (7.0 dB) indicates that the waveform branch adds architectural complexity without proportional benefit at this training scale. This is consistent with the broader finding that 100 training tracks are insufficient to fully train a waveform branch: the same data constraint that collapses a standalone Conv-TasNet to -1.1 dB likely limits the contribution of the waveform path within the hybrid as well. Hybrid fusion may yield larger gains with more training data or with a pretrained waveform branch fine-tuned jointly.

The spectrogram recurrent model reaches 6.5 dB, slightly below the spectrogram SI-SDR baseline (6.8 dB). Adding a bidirectional GRU at the U-Net bottleneck does not reliably improve performance, suggesting that 6-second training chunks do not expose the long-range temporal dependencies that recurrent processing is designed to capture. The additional parameters may also introduce mild overfitting given the limited training set. The Music Demixing Extended network (MDX) model reaches 5.6 dB, below the spectrogram combined baseline (5.8 dB), indicating that replacing U-Net convolutions with TDF blocks does not improve separation quality at this scale and training budget.

Computational Cost and Training Efficiency

Training time varies substantially across model families and reflects both architectural complexity and input representation. Spectrogram U-Net models require the longest runs: the magnitude L1 and SI-SDR conditions each train for approximately 5.3 to 5.5 hours (60 and 56 epochs respectively at roughly 5.5 minutes per epoch), and the spectrogram recurrent model extends to 6.1 hours over 60 epochs due to the additional BiGRU parameters at the bottleneck. The MDX model trains for approximately 1.8 hours over 30 epochs. By contrast, waveform Conv-TasNet models train in 35 to 45 minutes across all three loss conditions, roughly 8 to 10 times faster than their spectrogram counterparts. This speed advantage stems directly from the down sampled input: at 16 kHz with 2-second crops and batch size 4, each waveform training

step processes far fewer samples than a spectrogram model operating at 44.1 kHz with 6-second crops and batch size 8.

The efficiency gap inverts the cost-benefit relationship. Under the constraints of this project, waveform models train 10 times faster but produce results more than 8 dB worse. This gap is specific to the 16 kHz, 2-second crop configuration, not an inherent property of time-domain separation. Spectrogram models take substantially longer but converge reliably to competitive performance. For a fixed compute budget, the spectrogram approach delivers far greater return: the 5.5-hour spectrogram mag L1 run achieves 7.0 dB, while the combined waveform runs totaling under 3 hours across all three loss conditions peak at -1.07 dB. Training speed alone is a poor proxy for model efficiency. The relevant metric is SDR per unit of compute, and spectrogram models outperform waveform models by a wide margin on that measure given the constraints of this project.

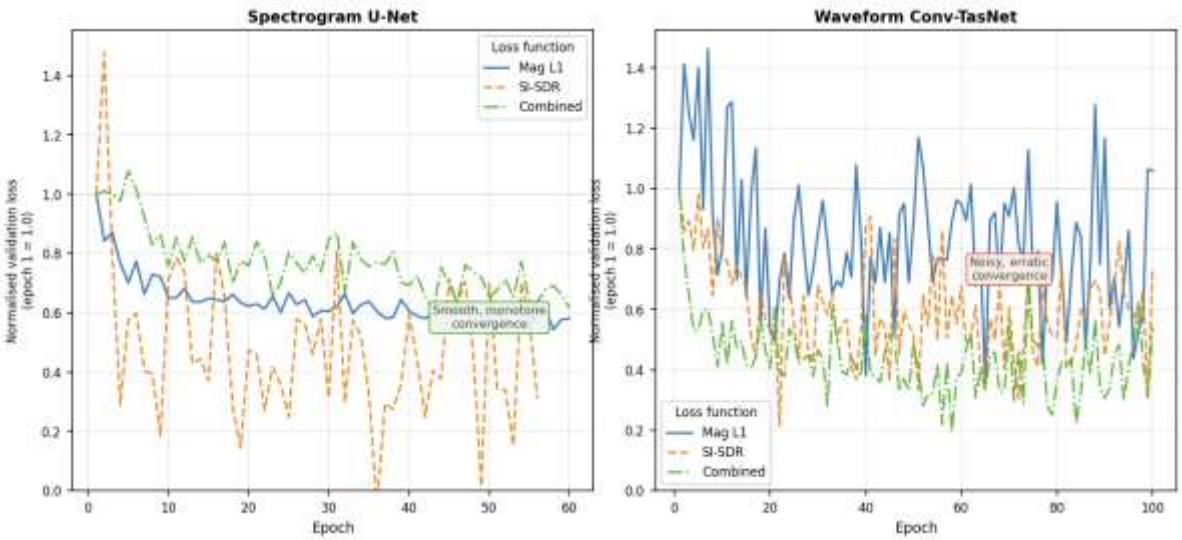

Figure 12. Normalized validation loss per epoch for the loss function ablation (spectrogram U-Net, left; waveform Conv-TasNet, right). Spectrogram models converge smoothly under all three loss conditions, reaching their best validation loss within 60 epochs. Waveform validation loss oscillates throughout training across all three conditions, reflecting the difficulty of optimizing a time-domain model on 100 training tracks with 2-second crops at 16 kHz.

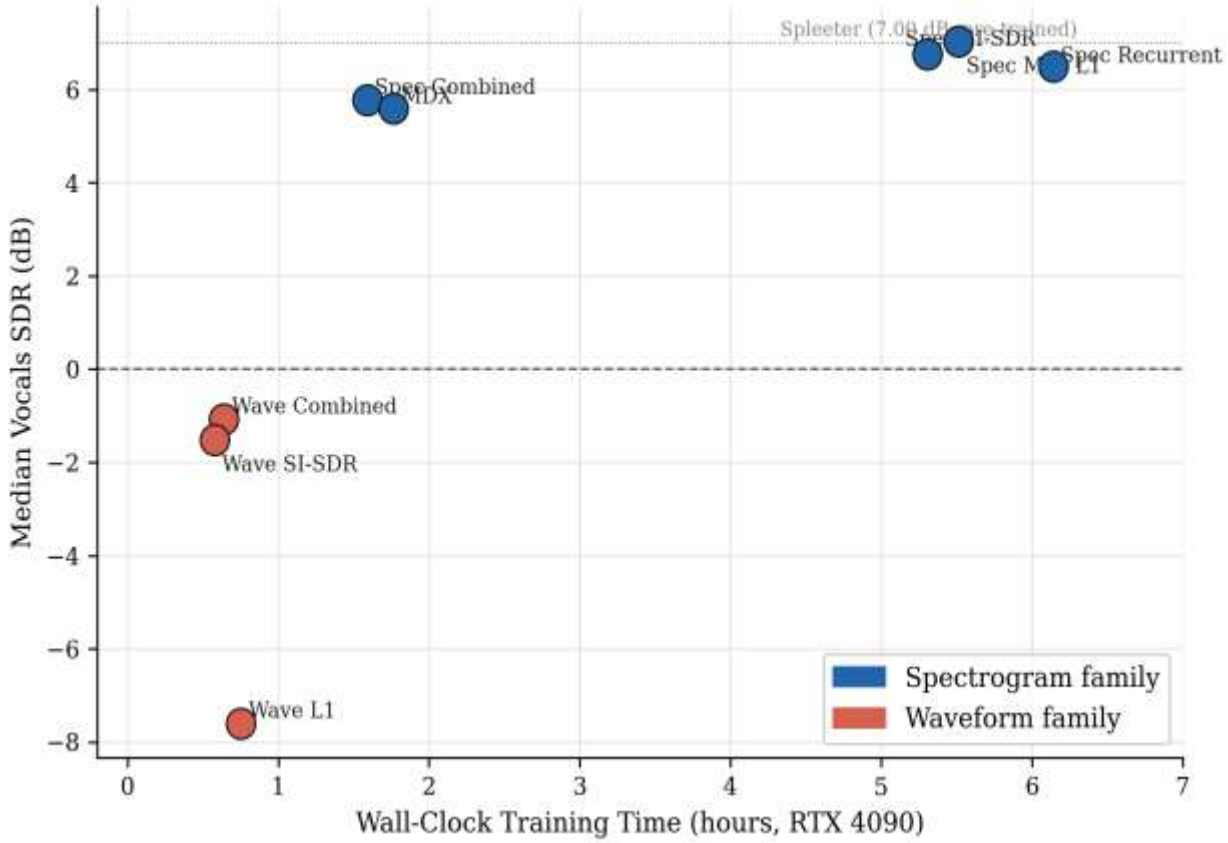

Figure 13. Wall-clock training time versus median vocals SDR for all trained models. Spectrogram models require 1.6–6.1 hours of training but achieve 5.6–7.0 dB SDR. Waveform models train in 35–45 minutes but reach at most −1.1 dB, illustrating that faster convergence at 16 kHz with 2-second crops does not compensate for the inductive bias advantage of spectrogram masking.

Per-Track Variance and Failure Cases

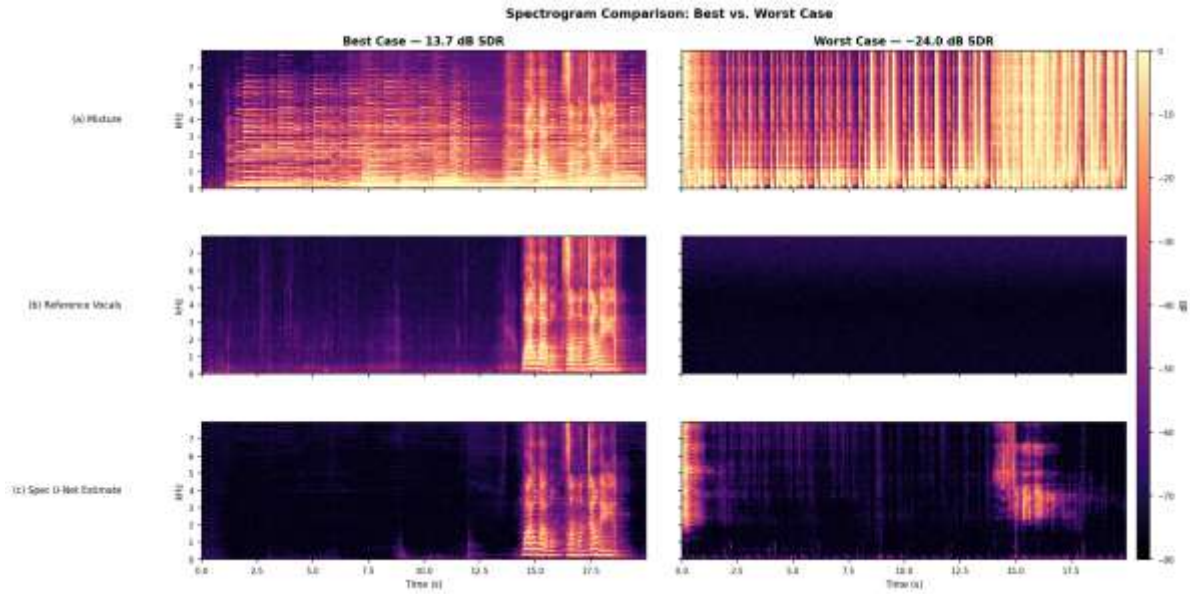

Figure 16. Spectrogram comparison for the best-performing and worst-performing test tracks (Spectrogram U-Net, magnitude L1 loss). Left: Lyndsey Ollard — Catching Up (13.7 dB SDR); Right: PR — Happy Daze (−24.0 dB SDR). Rows show (a) the stereo mixture, (b) the reference vocal stem, and (c) the model estimate. The best case has a sparse arrangement with clear vocal onset structure; the model estimate closely tracks the reference. The worst case has a fully saturated mixture; the model assigns accompaniment energy to the vocal channel.

Per-track SDR varies substantially even for the best model, from 13.7 dB (Lyndsey Ollard - Catching Up) to -23.9 dB (PR - Happy Daze). The striking finding is that this ranking is consistent across all model families: the same tracks that fail for the spectrogram U-Net also fail for the hybrid and waveform models, and at nearly identical magnitudes. Per-track SDR scores for all 50 test tracks and all models are reported in the Appendix A. PR - Happy Daze scores -23.9 dB (Spec Mag L1), -24.2 dB (Hybrid Combined), and -17.4 dB (Waveform Combined). Skelpolu - Resurrection scores between -10.70 and -16.4 dB across all models. This cross-model consistency indicates that failure is driven by acoustic properties of the source material rather than by any particular architectural limitation.

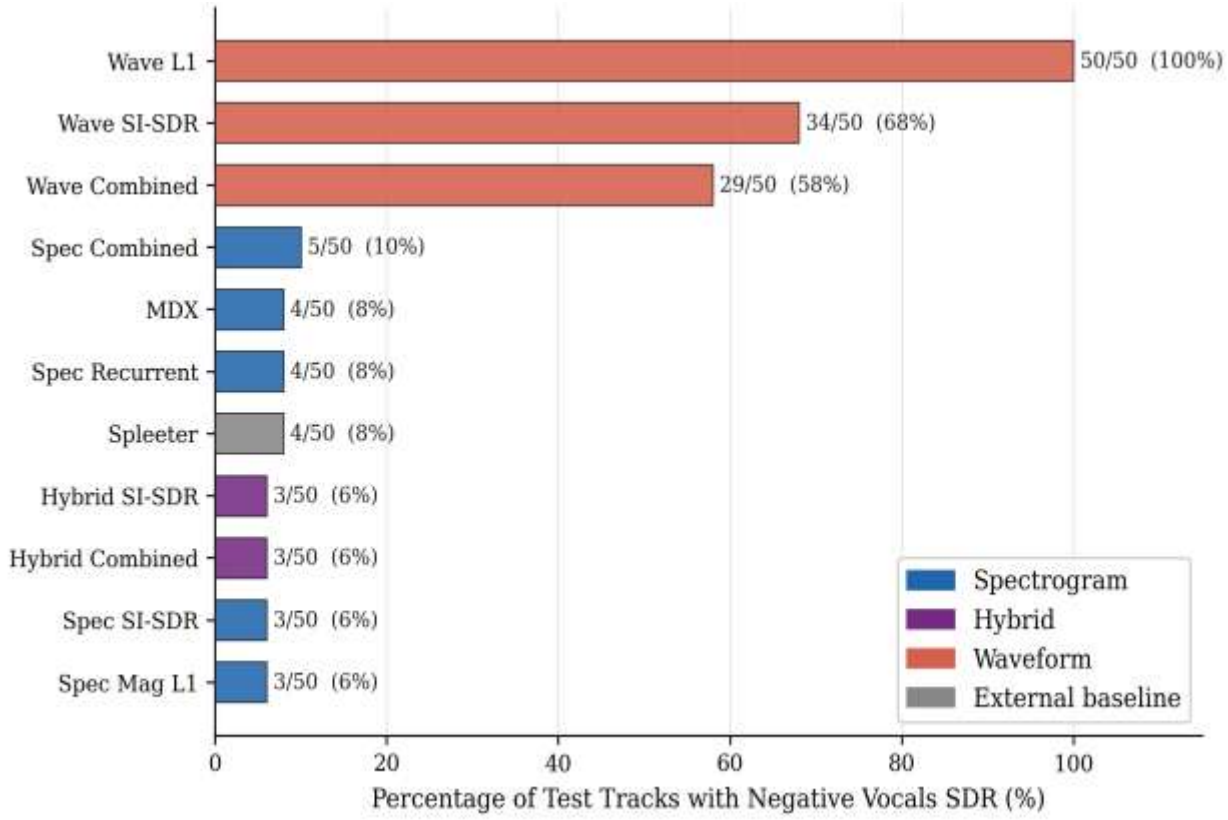

Figure 14. Proportion of the 50 MUSDB18-HQ test tracks on which each model produces negative vocals SDR (worse than silence). Spectrogram and hybrid models fail on only 3–5 tracks (6–10%); waveform models fail on 29–100% of tracks, with the L1 condition failing on every track in the test set.

Three acoustic conditions explain the consistent failures. First, percussive bleed: hi-hats and cymbals concentrate energy in the 4 to 12 kHz range, exactly the band that carries vocal sibilants and consonants. No frequency-domain mask can separate energy in a bin where two sources arrive simultaneously with similar magnitude. Second, distorted guitars and synthesizers spread dense harmonic series across 200 to 5000 Hz, the same range as vowel formants. When the harmonic structure of accompaniment occupies the same spectral bins as the vocal, the mixture spectrogram contains no local evidence to assign those bins to either source. Third, heavy reverb and room acoustics smear source energy across time frames, erasing the onset and offset cues that help the model distinguish vocal activity from background. The extreme result for PR - Happy Daze (-23.9 dB for the best model) suggests a near-inversion of the target signal: the model may have assigned the vocal spectral regions to the accompaniment estimate, which is consistent with a track where accompaniment dominates and the vocal is spectrally masked throughout.

The best-performing tracks share the inverse characteristics: sparse arrangements (Lyndsey Ollard - Catching Up is a solo acoustic recording), or a vocal timbre that is spectrally and culturally distinct from Western rock instrumentation (Sambasevam Shanmugam - Kaathaadi features Carnatic classical vocals with a timbral profile far outside the frequency bands dominated by drums and electric guitars). Waveform models are additionally sensitive to timbral distribution shift: the Conv-TasNet encoder learns basis functions tuned to the average training mixture, so tracks with unusual production styles or non-Western instrumentation fall outside the encoder's learned representation, amplifying the degradation seen in the worst cases.

These spectral failure modes produce characteristic perceptual artifacts. Estimates from the worst-performing tracks have a muffled, underwater quality caused by the model attenuating energy in contested frequency bins rather than assigning it confidently to either source. Waveform model outputs, particularly under L1 loss, additionally exhibit a hollow, low-energy timbre: the model has learned to suppress output amplitude globally rather than separate sources, producing something closer to silence than vocals. On moderately difficult tracks, spectrogram model estimates show metallic or phasey coloration at transients, an artifact of multiplying a predicted mask against the mixture magnitude and reusing the mixture phase at inference. These artifacts are perceptually distinct and identifiable by ear. A MUSHRA (MUltiple Stimuli with Hidden Reference and Anchor) listening study would establish whether the SDR ranking translates to consistent listener preference, and is a natural direction for follow-on evaluation.

Comparison to Spleeter

The spectrogram U-Net with magnitude L1 loss achieves 7.0 dB median vocals SDR, matching Spleeter at 7.0 dB on the same 50 test tracks. This result is notable because Spleeter (Hennequin et al. 2020) is an open-source, production-grade system whose pre-trained weights were trained on Deezer's internal music catalog, a large proprietary dataset unavailable to the research community, while the spectrogram U-Net is trained only on 100 MUSDB18-HQ tracks. The match reflects the effectiveness of the spectrogram masking approach rather than model sophistication: given a well-matched architecture and loss function, a simple U-Net reaches the same aggregate SDR as a heavily engineered external baseline. However, Spleeter achieves a higher first quartile (Q1 = 4.5 dB versus 4.1 dB), indicating better robustness on harder tracks. The spectrogram U-Net degrades more sharply on failure cases, suggesting that Spleeter's larger training set and architectural refinements provide more consistent performance across

diverse musical styles. The result also aligns with published spectrogram baselines: Open-Unmix (Stöter et al. 2019), trained on the same 100-track MUSDB18 training set, achieves 6.3 dB on MUSDB18, confirming that 7.0 dB represents a strong outcome for a spectrogram model at this data scale.

Conclusions

These results demonstrate that spectrogram-domain source separation is both reliable and data-efficient at the scale of the MUSDB18-HQ training set. A five-layer U-Net trained on 100 tracks with a simple magnitude L1 loss achieves 7.0 dB median vocals signal-to-distortion ratio (SDR) on the official 50-track test set, matching Spleeter, a production system trained on a large proprietary dataset, on the same evaluation. The result validates the spectrogram masking approach as a strong baseline that requires no architectural sophistication beyond a standard encoder-decoder and a well-chosen loss function.

The most consequential finding is that inductive bias matters more than model capacity at this data scale. The spectrogram U-Net constrains its output to a filtered version of the input mixture, a physically grounded prior that prevents hallucination and stabilizes training on limited data. The waveform Conv-TasNet carries no such constraint and must learn the full generative structure of vocal audio from scratch, a task that published results show is achievable given sufficient data and compute (Défossez et al. 2019), but that exceeds what a 100-track training set on consumer hardware can support. The -1.1 dB waveform result is therefore not a verdict on time-domain separation as a paradigm, but a precise illustration of where the inductive bias advantage of spectrogram masking becomes decisive.

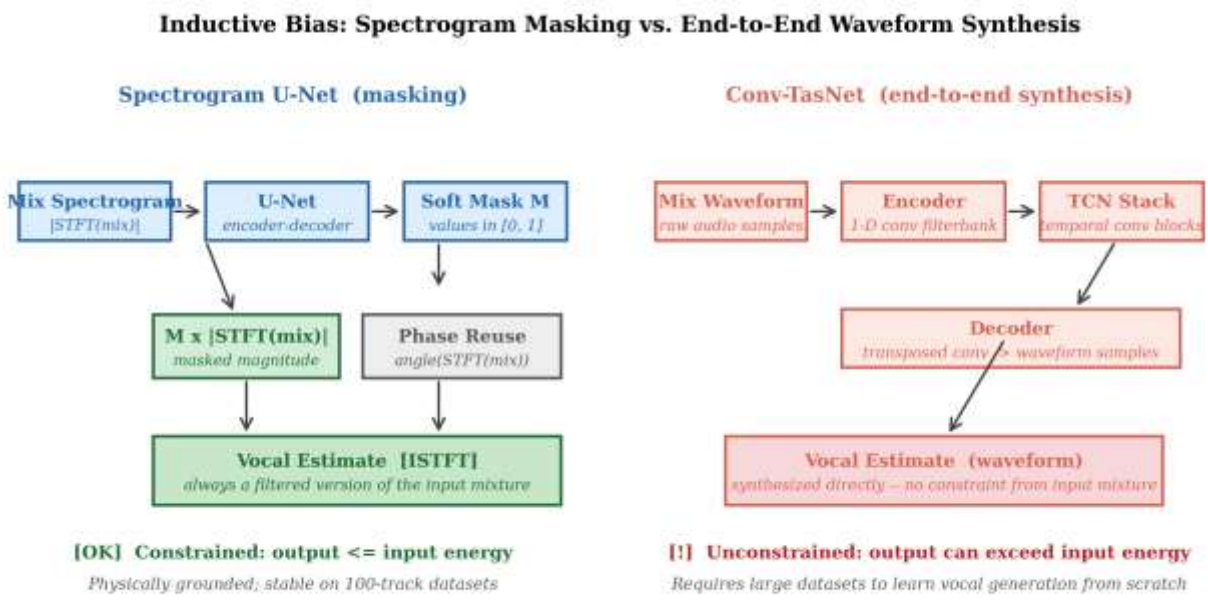

Figure 15. Architecture-level inductive bias comparison. Left: the spectrogram U-Net applies a soft mask M∈[0,1] to the input magnitude, constraining the output to a filtered version of the mixture. Right: Conv-TasNet synthesizes a waveform directly from the encoder representation with no constraint on output energy. The unconstrained path requires substantially more training data to learn stable vocal generation.

Two unexpected findings shaped the final design more than any architectural choice. The first was the waveform memory failure: an initial Conv-TasNet configuration at native 44.1 kHz and 6-second crops exhausted 24 GB of graphics processing unit (GPU) memory before producing a stable training run, forcing a reduction to 16 kHz and 2-second crops. This constraint became the single largest factor in the waveform performance gap. The second was the waveform L1 collapse: applying a loss function borrowed directly from the spectrogram pipeline to a time-domain model produced -7.6 dB median SDR, worse than the mixture input on most tracks. This failure clarified that architectural assumptions are embedded in loss functions as much as in network structure, a lesson that becomes immediately apparent when the model converges to silence.

Loss function selection is a design decision with architecture-specific consequences. Magnitude L1 loss is optimal for spectrogram masking because it directly minimizes the quantity the model predicts. Applying raw L1 loss to time-domain waveforms produces catastrophic failure (-7.6 dB) because sample-level amplitude differences are not meaningful targets for a source separator. This distinction generalizes beyond the configurations tested here: any loss function

must be matched to the output representation of the model, not borrowed from a related but structurally different architecture.

The per-track results also expose a consistency problem none of these models resolve. Tracks with dense, distorted, or heavily processed mixes produce negative SDR across every architecture tested, from spectrogram masking to hybrid time-frequency processing. The ceiling for vocal separation on these tracks is set by the acoustic properties of the source material, not by the choice of model. Closing that gap requires either richer training data that covers adversarial mix conditions, or separation objectives that explicitly account for perceptual quality rather than signal reconstruction fidelity. Both directions remain open problems in the field. The contribution of this work is a controlled comparison that isolates where each direction breaks down and why, providing a replicable foundation for future work at larger scale.

Directions for Future Work

Several directions emerge from these results for extending and improving vocal source separation beyond the configurations evaluated here.

- The most direct path to higher performance is more training data. The 100-track MUSDB18-HQ training set is the primary limiting factor for all models evaluated, and training on larger corpora such as MedleyDB, Slakh2100, or synthetically generated mixtures would be expected to yield substantial gains without any architectural change. Data augmentation offers a lower-cost alternative: pitch shifting and tempo stretching have been shown to contribute approximately 0.5 dB on MUSDB18 (Défossez et al. 2019) without modifying the model.
- Two architectural directions remain constrained by the hardware available for this work. The waveform Conv-TasNet was reduced to 16 kHz to fit within 24 GB video random access memory (VRAM). Retraining at 44.1 kHz using gradient checkpointing and mixed-precision inference would remove the most significant known disadvantage of the waveform approach and allow a direct comparison with spectrogram models at native sample rate. Beyond the convolutional baselines, transformer-based architectures such as Sepformer and TF-GridNet have recently set state-of-the-art results on MUSDB18 by replacing U-Net skip connections with self-attention layers that capture long-range temporal dependencies. The spectrogram stems model trained in this work also separates all four MUSDB18 stems but was not fully evaluated here; extending to four-

stem signal-to-distortion ratio (SDR) reporting would enable direct comparison with published multi-source systems.

- All models optimize signal reconstruction metrics that do not directly correspond to perceived audio quality. Incorporating perceptual loss functions such as multi-scale short-time Fourier transform (STFT) loss or pre-trained audio feature matching would better align the training objective with subjective quality. A complementary step is formal listening evaluation: a MUSHRA (MUltiple Stimuli with Hidden Reference and Anchor) study on the best and worst-performing tracks would establish whether SDR gains translate to audible improvements and provide ground truth for the failure cases identified in this work.

Data Availability

All experiments use the MUSDB18-HQ dataset, an uncompressed version of MUSDB18 containing 150 full-length stereo music tracks with isolated stems. The dataset is publicly available at https://doi.org/10.5281/zenodo.3338373 (Rafii et al. 2019). No proprietary or restricted data were used in this work.

Code Availability

The code developed for this project, including model implementations, training scripts, evaluation pipelines, and experiment configurations, is publicly available on GitHub at https://github.com/jeremycruzz/vocal-isolation. The repository includes all source code under src/, training and evaluation scripts under scripts/, and configuration files under configs/. A live interactive demo of the best-performing spectrogram model is hosted at jeremycruz.com.

Use of AI in Developing This Work

Claude (Anthropic), accessed via the Claude Code command-line interface, assisted with report drafting, section editing, document formatting, and structured review sessions. Specific contributions include reference list formatting, Introduction restructuring, Abstract revision, and figure integration. Cursor assisted with code debugging during model development. All experimental design, model training, evaluation, and interpretation of results are the author's own work.

Acknowledgements

I want to thank Dr. Alianna J. Maren for guidance, instructional resources, and the professional writing framework developed at Themesis, Inc. (themesis.com), which shaped the structure and presentation of this work. The MUSDB18-HQ dataset was created by Zafar Rafii, Antoine Liutkus, Fabian-Robert Stöter, Stylianos Ioannis Mimilakis, and Rachel Bitteur, and is gratefully acknowledged as the foundation for all experiments reported here.

References

Défossez, Alexandre, Nicolas Usunier, Léon Bottou, and Francis Bach. 2019. "Music Source Separation in the Waveform Domain." arXiv:1911.13254. https://arxiv.org/abs/1911.13254.

Hennequin, Romain, Anis Khlif, Felix Voituret, and Manuel Moussallam. 2020. "Spleeter: A Fast and Efficient Music Source Separation Tool with Pre-Trained Models." Journal of Open Source Software 5 (50): 2154. https://doi.org/10.21105/joss.02154.

Stöter, Fabian-Robert, Stefan Uhlich, Antoine Liutkus, and Yuki Mitsufuji. 2019. "Open-Unmix: A Reference Implementation for Music Source Separation." Journal of Open Source Software 4 (41): 1667. https://doi.org/10.21105/joss.01667.

Rafii, Zafar, Antoine Liutkus, Fabian-Robert Stöter, Stylianos Ioannis Mimilakis, and Rachel Bitteur. 2019. "MUSDB18-HQ: An Uncompressed Version of MUSDB18." Zenodo. https://doi.org/10.5281/zenodo.3338373.

Luo, Yi, and Nima Mesgarani. 2019. "Conv-TasNet: Surpassing Ideal Time-Frequency Magnitude Masking for Speech Separation." IEEE/ACM Transactions on Audio, Speech, and Language Processing 27 (8): 1256–66. https://doi.org/10.1109/TASLP.2019.2915167.

Jansson, Andreas, Eric Humphrey, Nicola Montecchio, Rachel Bitteur, Aparna Kumar, and Tillman Weyde. 2017. "Singing Voice Separation with Deep U-Net Convolutional Networks." In Proceedings of the 18th International Society for Music Information Retrieval Conference (ISMIR), 745–51.

Défossez, Alexandre. 2021. "Hybrid Spectrogram and Waveform Source Separation." arXiv:2111.03600. https://arxiv.org/abs/2111.03600.

Stöter, Fabian-Robert, Antoine Liutkus, and Nobutaka Ito. 2018. "The 2018 Signal Separation Evaluation Campaign." In Proceedings of the 14th International Conference on Latent Variable Analysis and Signal Separation (LVA/ICA), 293–305. https://doi.org/10.1007/978-3-319-93764-9_28.

International Telecommunication Union. 2015. “Method for the Subjective Assessment of Intermediate Quality Level of Audio Systems.” Recommendation ITU-R BS.1534-3. Geneva: ITU. https://www.itu.int/rec/R-REC-BS.1534/en.

List of Abbreviations

The following abbreviations are used in this report. Each is defined on first use in each major section.

Abbreviation	Definition
dB	Decibels
FFT	Fast Fourier Transform
GPU	Graphics Processing Unit
ISTFT	Inverse Short-Time Fourier Transform
MDX	Music Demixing Extended network
MUSDB18-HQ	Multi-Track Music Dataset (High Quality)
MUSHRA	MUltiple Stimuli with Hidden Reference and Anchor
PIT	Permutation-Invariant Training
SDR	Signal-to-Distortion Ratio
SI-SDR	Scale-Invariant Signal-to-Distortion Ratio
STFT	Short-Time Fourier Transform
TCN	Temporal Convolutional Network
VRAM	Video Random Access Memory

Appendix A: Hyperparameter Summary

Training Hyperparameters by Model Family

Parameter	Spectrogram U-Net	Conv-TasNet (Waveform)	Hybrid
Sample rate	44,100 Hz	16,000 Hz	44,100 Hz
Crop length	6.0 s	2.0 s	4.0 s
Batch size	8	4	6
Optimizer	Adam	Adam	Adam
Learning rate	1×10^{-3}	3×10^{-4}	1×10^{-3}
Grad clipping	None	5.0	None
Max epochs	60	60	60
Early stopping	20 epochs	20 epochs	20 epochs
GPU	RTX 4090	RTX 4090	RTX 4090

Table A1. Training hyperparameters for each model family. All models use the Adam optimizer with early stopping (patience 20 epochs) on a single NVIDIA RTX 4090. Conv-TasNet uses a reduced sample rate and crop length to fit within GPU memory constraints.

Appendix B: Individual Track Results

Table A1a. Per-Track Vocals SDR (dB) — Tracks 1-25 (best to worst)

Track	Spleeter	Spec Mag L1	Spec SI-SDR	Spec Combined	Spec Recurrent	MDX	Hybrid Combined	Hybrid SI-SDR	Wave Combined	Wave SI-SDR	Wave L1	Track Avg
MEDIAN	+7.0	+7.0	+6.8	+5.8	+6.5	+5.6	+6.3	+5.9	-1.1	-1.5	-7.6	+3.7
MEAN	+6.2	+5.8	+5.7	+4.1	+4.9	+4.7	+5.1	+4.9	-1.8	-2.3	-8.2	+2.6
Lyndsey Ollard - Catching Up	+13.9	+13.9	+13.8	+13.2	+13.8	+13.4	+13.7	+13.2	+6.8	+5.3	-5.0	+10.5
Sambasevam Shanmugam - Kast...	+12.3	+12.1	+11.2	+9.6	+11.4	+9.9	+11.2	+10.7	+3.0	+2.8	-4.2	+8.1
Side Effects Project - Sing...	+10.8	+10.8	+10.8	+9.1	+10.6	+9.2	+10.6	+10.6	+5.4	+4.3	-5.1	+7.7
The Sunshine Garcia Band - ...	+9.9	+9.8	+10.0	+9.3	+10.2	+9.5	+9.9	+9.1	+5.0	+4.0	-5.8	+7.4
Signe Jakobsen - What Have ...	+9.4	+10.1	+9.9	+8.9	+9.4	+9.0	+9.4	+9.0	+3.8	+2.1	-6.4	+6.8
BKS - Too Much	+10.7	+11.1	+10.7	+9.1	+9.6	+8.6	+9.8	+9.6	+0.6	+0.1	-6.7	+6.6
Moonmusic - Big Dummy Shake	+9.7	+9.9	+10.1	+9.0	+9.7	+8.6	+9.7	+9.0	+0.6	+0.2	-7.6	+6.3
AM Contra - Heart Peripheral	+10.8	+9.1	+10.1	+7.5	+9.2	+8.3	+9.4	+9.0	+0.7	+0.7	-6.6	+6.2
Enda Reilly - Cur An Long A...	+8.6	+8.7	+8.1	+7.5	+8.7	+8.5	+8.7	+8.4	+2.0	+2.0	-5.6	+6.1
The Mountaineering Club - M...	+10.4	+10.1	+10.2	+8.5	+10.2	+9.0	+10.1	+9.2	-1.3	-2.1	-8.7	+6.1
Tom McKenzie - Directions	+9.6	+9.6	+9.5	+7.7	+9.2	+7.9	+7.9	+8.2	+1.9	+1.1	-8.2	+5.9
Cristina Vane - So Easy	+9.1	+9.2	+8.8	+6.8	+7.6	+7.3	+7.5	+7.7	+2.4	+1.1	-6.9	+5.5
The Doppler Shift - Atrophy	+8.5	+8.6	+9.1	+7.4	+8.4	+7.9	+8.5	+8.2	+0.0	-0.9	-8.4	+5.2
Angels In Amplifiers - I'm ...	+9.0	+8.6	+8.0	+7.5	+7.1	+6.3	+7.2	+6.5	+1.2	-0.4	-6.5	+5.0
Trivul feat. The Fiend - W...	+8.3	+8.0	+7.2	+6.1	+6.9	+6.5	+7.0	+6.6	+1.4	+1.1	-5.9	+4.8
M.E.R.C. Music - Knockout	+7.6	+7.4	+6.7	+6.5	+6.9	+6.2	+6.6	+6.5	+1.1	+0.6	-6.6	+4.5
The Long Wait - Dark Horses	+7.5	+7.3	+7.0	+5.3	+6.6	+5.7	+6.6	+6.0	-0.1	-0.5	-4.5	+4.3
James Elder & Mark M Thomps...	+7.2	+6.8	+6.8	+5.9	+6.3	+5.7	+6.3	+6.1	+1.1	-0.3	-6.5	+4.1
Al James - Schoolboy Facina...	+6.5	+6.4	+6.0	+5.9	+5.2	+5.0	+5.4	+5.4	+2.9	+2.9	-4.0	+4.1
Mu - Too Bright	+7.1	+7.1	+6.9	+5.8	+6.8	+6.3	+6.4	+6.1	-0.1	-0.1	-7.3	+4.1
Nerve 9 - Pray For The Rain	+7.5	+7.6	+6.6	+6.7	+6.1	+5.3	+6.0	+5.9	+0.4	-0.4	-7.1	+4.0
The Easton Ellises (Baxmi) ...	+5.9	+6.2	+6.2	+5.0	+6.5	+6.2	+6.5	+6.4	+0.9	+0.3	-6.8	+3.9
Zeno - Signs	+6.9	+7.0	+7.3	+5.8	+6.6	+5.8	+6.6	+6.0	+0.0	-1.3	-7.6	+3.9
Juliet's Rescue - Heartbeats	+8.1	+7.4	+7.2	+6.0	+7.1	+6.2	+7.1	+6.6	-1.4	-2.4	-10.2	+3.8
Speak Softly - Like Horses	+6.3	+5.9	+6.5	+6.2	+6.5	+6.0	+6.3	+5.9	-1.9	-1.4	-5.0	+3.7

Table B1a. Per-track vocals SDR (dB): Tracks 1–25, sorted by average SDR (best to worst).

Per-Track Vocals SDR (dB) — Tracks 26–50, sorted by average SDR (best to worst)

Track	Spleeter	Spec Mag L1	Spec SI-SDR	Spec Combined	Spec Recurrent	MDX	Hybrid Combined	Hybrid SI-SDR	Wave Combined	Wave SI-SDR	Wave L1	Track Avg
MEDIAN	**+7.0**	**+7.0**	**+6.8**	**+5.8**	**+6.5**	**+5.6**	**+6.3**	**+5.9**	**-1.1**	**-1.5**	**-7.6**	**+3.7**
MEAN	**+6.2**	**+5.8**	**+5.7**	**+4.1**	**+4.9**	**+4.7**	**+5.1**	**+4.9**	**-1.8**	**-2.3**	**-8.2**	**+2.6**
Bobby Nobody - Stitch Up	+6.9	+7.1	+7.2	+5.4	+6.8	+5.6	+6.4	+6.0	-1.0	-1.3	-8.0	+3.7
Secretariat - Over The Top	+7.3	+7.9	+7.0	+6.8	+7.2	+5.6	+7.3	+6.4	-3.8	-3.5	-9.8	+3.5
Motor Tapes - Shore	+6.9	+7.3	+6.8	+6.2	+6.4	+5.4	+6.2	+5.4	-2.4	-2.9	-7.4	+3.5
Carlos Gonzalez - A Place F...	+6.2	+5.3	+5.4	+4.2	+4.9	+4.4	+5.1	+5.2	+0.8	+0.0	-5.0	+3.3
Arise - Run Run Run	+6.8	+5.4	+6.6	+4.3	+6.7	+4.5	+6.9	+4.8	-1.5	-1.6	-9.1	+3.1
Louis Cressy Band - Good Ti...	+7.9	+8.5	+6.4	+5.8	+5.5	+3.9	+4.9	+5.8	-2.7	-3.3	-10.0	+2.8
Hollow Ground - Ill Fate	+6.1	+6.4	+5.5	+5.1	+5.6	+4.5	+5.7	+4.1	-1.2	-2.9	-11.6	+2.5
Secretariat - Borderline	+6.7	+6.2	+5.3	+4.8	+5.4	+3.3	+4.9	+3.9	-2.5	-2.8	-8.6	+2.4
Little Chicago's Finest - M...	+10.3	+7.3	+8.3	+6.3	+8.4	+7.0	+8.3	+7.2	-18.3	-16.8	-5.5	+3.1
Forkupines - Semantics	+6.3	+6.0	+5.7	+3.4	+5.1	+4.5	+4.9	+4.8	-3.8	-4.3	-12.5	+1.8
Bultraker - Revo X	+2.8	+3.6	+3.9	+0.7	+3.9	+3.4	+3.5	+3.6	-1.5	-1.8	-4.8	+1.6
Ben Carrigan - We'll Talk A...	+4.2	+4.2	+4.0	+3.1	+4.4	+3.6	+4.2	+3.6	-2.0	-2.8	-9.3	+1.6
The Easton Ellises - Falcon...	+4.4	+2.4	+3.6	-1.1	+3.0	+3.4	+3.4	+3.5	-0.8	-1.9	-8.1	+1.1
Girls Under Glass - We Feel...	+4.8	+4.9	+4.8	+2.8	+4.4	+3.3	+4.1	+4.3	-5.0	-5.7	-12.2	+0.9
Raft Monk - Tiring	+3.7	+1.7	+2.5	-0.1	+3.5	+3.0	+3.0	+3.0	-1.5	-2.0	-9.1	+0.7
Detsky Sad - Walkie Talkie	+4.6	+3.8	+3.4	+3.9	+4.2	+2.4	+4.1	+3.5	-5.1	-6.4	-11.3	+0.6
Punkdisco - Oral Hygiene	+7.5	+4.5	+3.9	+2.4	+2.2	+1.0	+1.3	+1.0	-2.7	-3.0	-12.5	+0.5
Speak Softly - Broken Man	+3.7	+3.7	+2.4	+3.3	+3.3	+1.2	+1.9	+2.6	-8.0	-5.6	-4.8	+0.3
Georgia Wonder - Siren	+4.3	+2.8	+1.9	+2.5	+2.6	+1.8	+2.2	+2.3	-3.5	-4.2	-9.3	+0.3
Timboz - Pony	+3.3	+3.2	+1.7	+0.5	+3.5	+1.4	+3.2	+2.4	-2.4	-3.4	-11.3	+0.2
We Fell From The Sky - Not ...	+3.8	+3.5	+2.7	+0.3	+2.0	-0.1	+2.2	+1.5	-5.5	-6.3	-13.1	-0.8
PR - Oh No	-4.8	+0.9	+3.1	+1.1	-0.4	+1.0	+0.4	+4.1	-13.2	-13.0	-12.9	-3.1
BKS - Bulldozer	-0.2	-0.3	-2.5	-2.9	-3.4	-4.6	-3.4	-3.7	-8.3	-9.1	-14.7	-4.8
Skelpolu - Resurrection	-9.1	-10.7	-16.4	-19.7	-20.7	-9.5	-13.6	-17.3	-14.2	-12.7	-15.6	-15.0
PR - Happy Daze	-14.4	-24.0	-15.7	-17.3	-20.1	-12.1	-24.2	-15.5	-17.4	-16.6	-12.2	-19.4

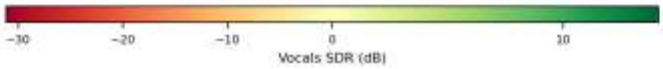

Table B1b. Per-track vocals SDR (dB): Tracks 26–50, sorted by average SDR (best to worst).

Appendix C: Log of AI Interactions

The following log records the AI-assisted tasks performed during this project using Claude (Anthropic), accessed through the Claude Code command-line interface. All experimental design, model training, and evaluation were performed independently. AI assistance was used for report writing, figure generation, and analysis scripting.

Phase	Tool	Date	Description
Code Development	Cursor	Feb 2026	AI-assisted debugging of model training scripts, data loading pipelines, and evaluation code.
Report Writing	Claude (Claude Code)	Mar 2026	Restructured Introduction: moved Key Contributions to paragraph 2, added formatted contribution box, tightened Abstract opening.
Report Writing	Claude (Claude Code)	Mar 2026	Formatted reference list with Chicago-style hanging indents.
Figure Generation	Claude (Claude Code)	Mar 2026	Generated and integrated figures including separation pipeline diagram and per-track SDR distribution plots.
Report Writing	Claude (Claude Code)	Mar 2026	Added per-section acronym definitions throughout (SDR, SI-SDR, STFT, TCN, GPU, VRAM, FFT, ISTFT, MDX, PIT). Added List of Abbreviations appendix.